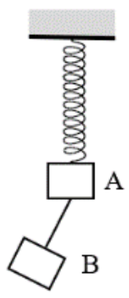
1. An object moves with velocity **v** and a constant acceleration **a**. Which of the following expressions is also constant (but not zero)?

   (A) $\frac{\mathrm{d}|\mathbf{v}|}{\mathrm{d}t}$ (B) $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathbf{v}}{v}\right)$ (C) $\frac{\mathrm{d}}{\mathrm{d}t}(v^2)$ (D) $\left|\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\right|$

2. A block A is connected to the ceiling (天花板) through a spring, and is restricted that it can only move vertically. Another block B is connected to it with an inextensible (不可伸长) string, and can swing freely. The volume of A, B is neglected. How many degrees of freedom does the system have?



   (A) 2 (B) 3 (C) 4 (D) 5

3. Which of the following statements is NOT true?

   (A) Gravitational potential is reversely proportional (反比) to distance.
   (B) A 1-dimensional force must be conservative if it is a function of position.
   (C) The energy of a simple harmony oscillator is proportional to the (amplitude)^2
   (D) The angular frequency of a specific simple harmony oscillator is constant.

4. Which of the statements about rotation of a rigid body is TRUE?

   (A) A net torque exerted on a rigid body must alter its kinetic energy.
   (B) A heavier rigid body must have a greater moment of inertia.
   (C) Kinetic energy of rotation is proportional to the square of angular momentum. $\frac{L^2}{2I}$
   (D) For a specific point on the rigid body, the magnitude of acceleration of it is proportional to the whole rigid body.

5. Which of the statements about electric field is NOT true?

   (A) The curl of an electric field is provided by the induced electric field. $\nabla \times E = \frac{\partial B}{\partial t}$
   (B) Electric field must be perpendicular to the surface of a conductor in case of electrostatic equilibrium.
   (C) The electric field flux is zero on a surface where the electric field vector is always along it.
   (D) 必须由非电磁力产生 ×
   (D) Electromotive force must be generated by non-electromagnetic interactions.
   不能由静电力 electrostatic，应由电磁力产生

6. Which of the statements about magnetic vector potential is TRUE?

   (A) A changing magnetic vector potential generates electric field.
   (B) Difference of magnetic vector potential is the integral of magnetic field.
   (C) The divergency of magnetic vector potential must be zero.
   (D) Magnetic vector potential cannot be zero where the magnetic field is not zero.

7. A particle of mass m=2kg is driven by a force varying with time: $\mathbf{F} = \mathrm{e}^{-2t}\mathbf{i} - \sqrt{t}\mathbf{j} + 2\mathbf{k}$, Where **F** is in newton and t is in second. At t=0, the particle is at the origin x=y=z=0 (in meter), and the initial velocity is $\mathbf{v}_0 = -2\mathbf{i} + \mathbf{k}$ (in m/s). Determine the position, the speed and the power of the force at t=1.4s.

   $a = \frac{e^{-2t}}{2} i - \frac{\sqrt{t}}{2} j + k$ $v = v_0 + \int_0^t a\,dt$ $r = r_0 + \int_0^t v\,dt$

8. A 1-dimensional conservative force is defined in the interval 0.1<x<5:

$F=0,\quad \frac{\pi}{x}=\frac{\pi}{2}+k\pi=\frac{1+2k}{2}\pi,\quad x=\frac{2}{1+2k}$

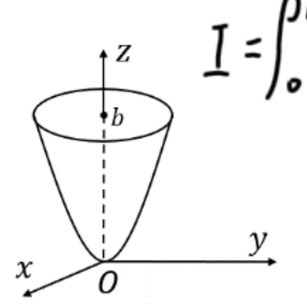
$x=2,\ \frac{2}{3},\ \frac{2}{5},\ \frac{2}{7},\ \frac{2}{9},\ \frac{2}{11},\ \frac{2}{13},\ \frac{2}{15},\ \frac{2}{17},\ \frac{2}{19}$

$-\frac{dU}{dx}=-\frac{\pi}{x^2}\sin\frac{\pi}{x}$

$$F=\cos\frac{\pi}{x}\quad (0.1<x<5)$$

Determine the positions of equilibria in the interval and their stabilities.

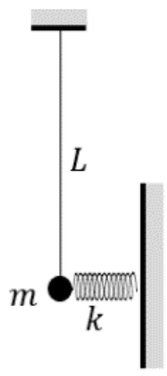
9. A uniform block of mass m takes over this region in the space: $x^2+y^2<z<b$, and b is a positive constant. Determine its moment of inertia about the z-axis. (**Hint**: you need to get its density by calculating its volume.) $V=\int_0^{2\pi}d\theta\int_0^{\sqrt{b}}r\,dr\int_{r^2}^{b}dz=\frac{\pi}{2}b^2$



$I=\int_0^{\sqrt{b}}r^2\cdot\frac{2m}{\pi b^2}\cdot 2\pi r(b-r^2)dr$

$m=\rho V=\frac{\pi}{2}b^2\cdot\rho,\quad \rho=\frac{2m}{\pi b^2}$

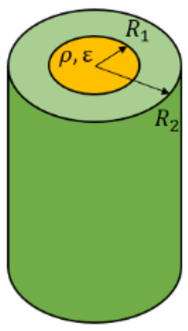
10. A simple pendulum of length L and mass m is connected to the wall through a horizontal spring, of which the stiffness coefficient is k, and the mass is neglected. The spring is relaxed when the pendulum is rest at the equilibrium. (a) Set the angular displacement of the pendulum θ as the general position coordinate, and give out the Lagrangian function, general momentum and Hamiltonian canonical equations of the system. (b) If the amplitude is not small (i.e we cannot consider that $\sin\theta=\theta$), but the spring is always horizontal by moving vertically, what is the Lagrangian function and general momentum? (a) $K=\frac{1}{2}mv^2=\frac{1}{2}mL^2\dot{\theta}^2$ $\quad L=K-U$



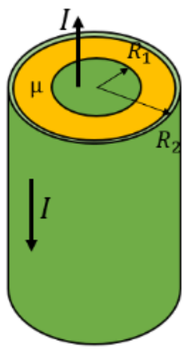
$\sin\theta=\theta$ $\quad U_1=\frac{1}{2}KA^2=\frac{1}{2}k(L\sin\theta)^2$ $\quad P=\frac{\partial L}{\partial\dot{\theta}}$

$\cos\theta=1-\frac{1}{2}\theta^2$ $\quad U_2=\frac{1}{2}mgL(1-\cos\theta)$

11. A long cylindrical insulator of radius R_1 carries a uniform charge density ρ, and is covered by a coaxial (共轴) conductor shell of radius R_2, which carries no net charge. The permittivity of the insulator is ε. (a) Determine the electric field distribution all over the space. (b) Assume that the electrostatic potential is zero on the central axis of the system, determine the potential where the distance to the axis is R_3 ($R_3>R_2$).



12. (Coaxial cable, 共轴电缆) A long cylindrical conductor wire of radius R_1 and a thin cylindrical conductor tube R_2 are placed coaxial, and carries uniform electric current I in opposite directions. The space between them is filled with insulator of permeability μ. Determine the magnetic field all over the space (not including the tube).

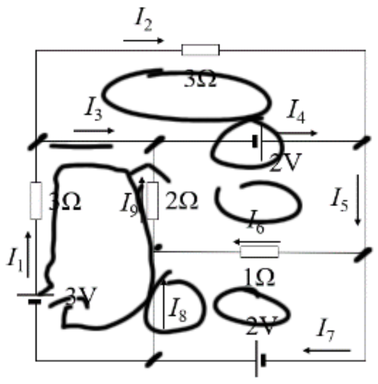


13. In a complex electromagnetic field, the electrostatic potential and magnetic vector potential is

$$V=x^2+3y-2yz-2t,\ \mathbf{A}_{\mathrm{M}}=2x\mathbf{i}-xz^2\mathbf{j}+3y^2t\mathbf{k}$$

where V is in volt, and $\mathbf{A}_{\mathrm{M}}$ is in tesla·meter. For the position x=-2, y=1, z=0 (in meter) and the time t=1 (in second), determine: (a) the electrostatic field $\mathbf{E}_s$; (b) the magnetic field $\mathbf{B}$; (c) the induced electric field $\mathbf{E}_i$; (d) the charge density ρ (assume that the permittivity is just ε_0 here). (e) the conducting current density $\mathbf{j}$ (assume that the permeability is just μ_0 here). (**Hint**: get the total current from the magnetic field, and get the displacement current density from the electric field.)

14. Write down all the equations about the electric circuit shown in the figure according to Kirchhoff's rules.



$I_1 = I_2 + I_3$

$I_4 = I_3 + I_9$

$I_5 = I_2 + I_4$

$I_5 = I_6 + I_7$

$I_9 = I_6 + I_8$

$I_7 = I_1 + I_8$

$3 = 3I_1 - 2I_9$

$2 = I_6 + 2I_9$

$2 = -I_6$

$2 = -3I_2$